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Applications of Computed Tomography (CT) in environmental soil and plant sciences

Huan Zhang [a], Hailong He [a,b,c,*], Yanjun Gao [d], Ahmed Mady [e], Vilim Filipović [f,g], Miles Dyck [h], Jialong Lv [a,b], Yang Liu [i,**]

[a] College of Natural Resources and Environment, Northwest A&F University, Yangling 712100, Shaanxi, China
[b] Key Laboratory of Plant Nutrition and the Agri-Environment in Northwest China (Ministry of Agriculture), Northwest A&F University, Yangling, Shaanxi 712100, China
[c] The State Key Laboratory of Soil Erosion and Dryland Farming on the Loess Plateau, Institute of Soil and Water Conservation, CAS&MWR, Northwest A&F University, Yangling, China
[d] Xi'an Key Laboratory of Cardiovascular and Cerebrovascular Diseases, Xi'an No.3 Hospital, the Affiliated Hospital of Northwest University, Xi'an 710000, Shaanxi, China
[e] Soil Science Department, Faculty of Agriculture, Ain Shams University, 11241 Cairo, Egypt
[f] Future Regions Research Centre, Geotechnical and Hydrogeological Engineering Research Group, Federation University, Gippsland, VIC 3841, Australia
[g] Department of Soil Amelioration, Division for Agroecology, Faculty of Agriculture, University of Zagreb, 10000 Zagreb, Croatia
[h] Department of Renewable Resources, University of Alberta, Edmonton, AB T6G 2H1, Canada
[i] Key Laboratory of Resource Biology and Biotechnology in Western China (Ministry of Education) and College of Life Science, Northwest University, Xi'an, Shaanxi 716000, China

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A B S T R A C T

Computed tomography (CT) in combination with advanced image processing can be used to non-invasively and non-destructively visualize complex interiors of living and non-living media in 2 and 3-dimensional space. In addition to medical applications, CT has also been widely used in soil and plant science for visual and quantitative descriptions of physical, chemical, and biological properties and processes. The technique has been used successfully on numerous applications. However, with a rapidly evolving CT technologies and expanding applications, a renewed review is desirable. Only a few attempts have been made to collate and review examples of CT applications involving the integrated field of soil and plant research in recent years. Therefore, the objectives of this work were to: (1) briefly introduce the basic principles of CT and image processing; (2) identify the research status and hot spots of CT using bibliometric analysis based on Web of Science literature over the past three decades; (3) provide an overall review of CT applications in soil science for measuring soil properties (e.g.,

Abbreviations: 2D, Two-dimensional; 3D, Three-dimensional; AFCM, Adaptive Fuzzy C-mean Method; AW, Amplitude-wavelength; BTCs, Breakthrough curves; CAS, Cation adsorption sites; CDE, Convective dispersive equation; CEC, Cation exchange capacity; CO_2, Carbon dioxide; CS, Silica-sol; CT, Computed tomography; DECT, Dual-energy CT; DEM, Digital element method; DLR, Deep-learning reconstruction algorithm; DRP, Digital Rock Physics; ERT, Electrical Resistance Tomography; EDS, Energy dispersive X-ray spectroscopy; FBP, Filtered back projection; FEGSEM-EDX, Field emission scanning electron microscopy coupled with microanalysis; FTCs, Freeze-thaw cycles; GAM, Gamma ray attenuation method; GV, Gray value; KDW, Kinematic dispersive wave; K_s, Saturated hydraulic conductivity values; KW, Kinematic wave; LB, Lattice Boltzmann; LSI, Loess structure index; MDLL, The mean diameter of the limiting layer; micro-XRF, Synchrotron micro-X-ray fluorescence; MOSAIC, Morphological model; MRI, Magnetic resonance imaging; MS, Microspheres; NCT, Neutron computed tomography; NMR, Nuclear Magnetic Resonance; OsO_4, Osmium tetroxide; OTO, Osmium-thiocarbohydrazide-osmium; PT-CT, Photon counting computed tomography; PTFs, Pedotransfer functions; PSD, Pore size distribution; RSA, Root system architecture; SART-TV, Simultaneous Algebraic Reconstruction Technique-Total Variation; SD, Standard deviation; SEM, Scanning Electron Microscope; SMAS, Soil Microstructure Analysis System; SIP, Spectral induced polarization; SNR, Signal- noise ratio; SOC, Soil organic carbon; SOM, Soil organic matter; SWRC, Soil water retention curve; SXRCT, Synchrotron X-ray computed tomography; SRXTM, Synchrotron Radiation X-ray Tomographic Microscopy; SR-μCT, Synchrotron Micro X-ray CT; TLS, Total link strength; TMM, Truncated multifractal method; TOF, Time-of-flight; Vis-NIR, Visible and near infrared spectroscopy; WDCs, Wetting and drying cycles; WoSCC, Web of Science Core Collection; XACT, X-ray induced acoustic computed tomography; XAS, X-ray absorption spectroscopy; X-ray μCT, X-ray microtomography; γ-ray CT, Gamma-ray CT.

* Corresponding author at: College of Natural Resources and Environment, Northwest A&F University, Yangling 712100, Shaanxi, China.
** Corresponding author.
E-mail addresses: ZH_2020@nwafu.edu.cn (H. Zhang), hailong.he@hotmail.com (H. He), nige.001@stu.xjtu.edu.cn (Y. Gao), ahmed_mady@agr.asu.edu.eg (A. Mady), vfilipovic@agr.hr (V. Filipović), miles.dyck@ualberta.ca (M. Dyck), ljlll@nwsuaf.edu.cn (J. Lv), liuyangent@nwu.edu.cn (Y. Liu).

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porous soil structure, soil components, soil biology, heat transfer, water flow, and solute transport); and (4) give an overview of applications of CT in plant science to detect morphological structures, plant material properties, and root-soil interaction. Moreover, the limitations of CT and image processing are discussed and future perspectives are given.

1. Introduction

Sustainable agriculture involves addressing the agricultural and environmental issues we are facing in the present. Soils supports about 95 % of agricultural production and is a critical resource on which we depend for survival (Borrelli et al., 2020). More than about 38 % of the world's arable land has been persecuted by soil degradation since 1945 (Liu et al., 2010), and reports on regional soil degradation in areas such as the European-Mediterranean region are clearly state the increasing degradation (Ferreira et al., 2022a). Soil pollution is another cause of degradation of valuable farmland resources, as the harmful elements may enter the food chain through crops and grains that can even damage human and animal health (Li et al., 2021b). Appropriate techniques are required to better understand the properties and processes of soils subjected to degradation or pollution and to provide optimal growing conditions for plants. There are numerous techniques available to measure soil and plant properties (He et al., 2021a, 2018, 2021b), but very few of them are non-destructive and non-invasive. In recent years, computed tomography (CT), which alleviates the need to disturb soil architecture, has become widely used (Baveye et al., 2022).

The non-destructive and non-invasive characteristic allows CT to perform visualization and quantitative analysis of the interior of porous structures (Haubitz et al., 1988). This feature makes CT scanning widely applicable in the field of earth and environmental research (Abdallah et al., 2021; Heijs et al., 1995; Ketcham and Carlson, 2001). In addition, CT is suitable for most materials (except for some very dense objects) and analysis can be performed regardless of the geometry of the object within the required dimensions. In addition, it requires little or no sample pre-treatment (Pierret et al., 2002), which makes it attractive and explains why it is becoming essential equipment in many disciplines (Baveye et al., 2022). However, it should also be noted that the access to CT scanner remains relatively limited for many institutions (related to the cost of the device and required expertise) and the accurate analysis of the resulting images is challenging and time consuming. The combination of CT scanning and advanced image processing techniques allows the sequential generation of a series of two-dimensional (2D) grayscale slice maps and the reconstruction of a complete three-dimensional (3D) sample, i.e., a stack of 2D slices (Fang et al., 2019a; Xiong et al., 2020; Zhang et al., 2019), allowing the construction of 3D models for numerical simulation that can perform seepage simulation, heat transfer simulation, etc. (Li et al., 2021a; Yu et al., 2019). It also means that the complex internal structure can be observed using CT in soils or plants, given that the finite resolution of the CT meets the research objectives.

These abovementioned characteristics (e.g., non-destructive and non-invasive nature, visualization of soil/plant structure, rapidity, combination with other techniques) expand the scope of research in soil and plant sciences using CT, making it suitable for studying complex pore networks of soils (Xu et al., 2021b; Zhang et al., 2021c), the diverse morphological structures of plants (Zhou et al., 2021), intersecting rhizosphere soils (Vetterlein et al., 2021), and hydraulic properties (Schluter et al., 2020). For instance, analyzing and interpreting the pyrolytic behavior of wooden species from a 3D perspective (Murai et al., 2021; Zhou et al., 2021), monitoring the formation and recovery of embolism of xylem (Secchi et al., 2021), visualizing the voids in plant reproductive organs (Xiao et al., 2021), and revealing grain traits in different crop species (Zhou et al., 2021) among others, CT scanning provides new insights into these studies that cannot be obtained via the traditional light or scanning electron microscopy analyses of thin sections. Moreover, 3D visualization using CT allows a more detailed characterization of the transport processes of solutes and environmental tracers in soils or plants (Ziegler-Rivera et al., 2021), based on which we can apply CT to trace and explain contaminant transport, spatial and temporal distribution characterization etc. For example, Gattullo et al. (2020) evaluated the chromium pollution of agricultural soils using CT and studied the distribution of chromium at different scales. Combing CT with fluorescence macrophotography, Soto-Gómez et al. (2018) found that the average pore surface is linearly related to the dispersion coefficient of bromide. Their results showed that the dimensionless microspheres (MS) transfer coefficient between the matrix and the macropores can well describe the retention percentage of colloidal fluorescent microspheres. Scotson et al. (2021a) used X-ray CT to trace the activity of soluble iodinated contrast agent to simulate solute migration in soil column. Their results indicated that plant roots are able to reduce the depth of solute penetration which was found to be correlated with the age of the plants. Kirk et al. (2019) used non-invasive CT scanning to image roots and bubbles in flooded soils and discussed the emission pathways of carbon dioxide (CO_2) generated from flooded rice fields in conjunction with mathematical models. They concluded that the flux of CO_2 through roots is equivalent to one-third of the daily photosynthetic fixation. These studies all have extraordinary implications for agri-environmental security.

Given the importance and wide application of CT, a detailed description of the principle and application of computerized axial tomography appeared as early as 1999 (Duliu, 1999). Carlson (2006) thoughtfully summarized and explained some professional terms that would appear during CT scanning for future generations. Probably limited by technology before the early years, their focus was mainly on rocks and soil particles. It is worth noting that while the application of X-ray CT to the study of granular geotechnical materials, coarse sands or glass beads is intrinsically interesting, the ability of CT to visualize complex soil structure and pore space is what needs to be explored as the soil contain much more than just mineral particles. Therefore, the information obtained in these idealized systems may deviate from the real soil (Baveye et al., 2018). These past studies had a limited understanding of the soil. Pires et al. (2010) provided the advancement of CT applications in soil physics, but they limit their review to Brazilian related applications and lack a critical discussion of these applied techniques. With the rapid development of CT technology, the application areas of CT are expanding, which brings new challenges to the measurement process and image processing of CT. This is especially true with the improved equipment and processing techniques for increasingly complex and in-depth research questions in the soil or plant research fields. In addition, although review articles have been published in recent years, they have focused on the technical progress in imaging plant roots and biological processes (Downie et al., 2015; Kumi et al., 2015; Piovesan et al., 2021) or the evolution of microstructures (Tang et al., 2022), with fewer case studies review of CT applications involving the integrated field of soil and plant research.

In this general context, the objective of this study is to present an updated review on CT applications in soil and plant sciences. The review starts with a brief introduction of the knowledge and background of CT and the basic principles of image processing. The important advances obtained in CT scanning in recent years are summarized and a series of targeted process parameters used for CT scanning and image processing in related directions are collated. The energy types and limitations of CT suitable for different directions are also discussed. In addition, bibliometrics is used to map the research trend and research hotspots for applications of CT. The example applications of CT in environmental soil and plant science in recent years are systematically reviewed. The review ends with the existing knowledge gaps and future prospects beeing discussed.

2. Basics of CT and image processing

2.1. Principles of CT

The working principle of commonly used CT is based on the fact that different objects/components have different absorption rates for permeable rays (Yan et al., 2021). CT can be based on X-ray CT, γ-ray CT, neutron CT, and X-ray microtomography (X-ray μCT) according to energy sources and resolution. A highly sensitive detector is used to capture the intensity of the rays attenuated by the sample, which is used as an analog signal input to process the data and reconstruct the image, or to reconstruct the details of a certain area with the help of mathematical algorithms. Different gray color levels are obtained in the reconstructed image depending on the material and geometry, enabling real-time observation and processing (Bonnet et al., 2003). Each slice of the CT image is composed of a certain number of pixels with different gray levels ranging from black to white. Note that radiation with low attenuation is shown in black, while high attenuation radiation is mostly assigned as white, with gray shading between black and white. For each pixel, the final display color of the pixel is determined by controlling the composite ratio of the three primary colors of R (red), G (green), and B (blue). Therefore, RGB (0 - 255) images can be converted to grayscale images (Panneton and Brouillard, 2009). These pixels of a slice are arranged in a matrix, whose gray levels provide information on the X-ray attenuation of the corresponding sample material at each point in that slice or the cross-section (Azeredo et al., 2013).

The scanning time of a CT device depends primarily on the sample (size and resolution) and rotation (degree of rotation, rotation step, average number of frames in a single rotation). There is a tradeoff between the sampling time and the quality of the grayscale image. To avoid detector saturation, the acquisition time is recommended to be in the range of 40 % - 60 % of the acquired camera brightness (Vaz et al., 2011). In general, the X-ray spectrum used in conventional CT scanners has a polychromatic characteristic. Most reconstruction methods cannot correctly check the nonlinearity of polychromatic X-rays (Cao et al., 2018), which causes beam hardening that can lead to artifacts and degrade the quality of the reconstructed image (Yang et al., 2020). Better peak separation is observed with the application of filters. Research has shown that the efficiency of common aluminum filters was about 10 % lower than filters made of materials such as copper, brass or iron (Jennings, 1988), but a higher level of experience and skill is required to select the right filter and hardware thickness (Romano et al., 2019). Conventional CT scanners always integrate all signals of detected transmitted X-rays into a single attenuated signal and do not record information about their respective energies (Si-Mohamed et al., 2017). Polychromatic CT based on photon-counting detectors works by acquiring several different energy data sets, which allow the energy threshold to be selected according to the application, and allow the detection of K-edges of a given element for more effective imaging of specific substances (Si-Mohamed et al., 2017). In addition, synchrotron X-ray beams are typically mono-chromatic, i.e., have a specific, tunable energy. Both polychromatic CT and synchrotron monochromatic X-ray beams can maximize signal-to-noise and have better spatial resolution while reducing the absorbed radiant energy (Melli et al., 2018). For energy sources, the commonly used energy sources are ionizing radiation (X or γ rays, neutrons), optical, electronic, and ultrasound etc., all of which are nondestructive but each with particular advantages (Yan et al., 2021). It is worth mentioning that the hyperspectral CT introduces spectral reconstruction and identification techniques to achieve a combination of structural and compositional analysis. It includes X-ray fluorescence spectroscopy (XFS), X-ray diffraction spectroscopy (XDS), X-ray absorption spectroscopy (XAS) (Fang et al., 2019b). X-ray induced acoustic computed tomography (XACT) and photon counting computed tomography systems (PT-CT) can be effective in reducing radiation damage (Boccalini et al., 2021; Flohr et al., 2020; Kiji et al., 2020; Maruhashi et al., 2019).

2.2. Development of CT

CT has evolved over multiple generations since its introduction. Although X-ray CT is very instrumental and widely used for nondestructive visualization and evaluation of the interior of porous materials, there are still uncertainties in many aspects of the measurements that require further calibration and optimization (Schmitt et al., 2018). The uncertainty in CT dimensional measurements can arise from different measurement parameters and conditions (e.g., detection system, dimensional resolution, radiation generation, and imaging processing) as well as from the material under investigation (Villarraga-Gomez et al., 2020), which requires calibration and optimization of parameters. For instance, Lin et al. (2019) calibrated the center-of-rotation parameters by an automatic calibration method based on sinusoidal symmetry. Villarraga-Gómez et al. (2020) described that the limitations of CT measurement range are primarily determined by the penetrating depth of X-rays through the materials. They argued that CT measurement performance could be improved by effectively setting CT parameter constraints in terms of resolution and measurement range. Villarraga-Gómez et al. (2021) further utilized ray projection and radon spatial analysis to simulate X-ray traversal paths and to find the optimal orientation of the estimated sample, i.e., the tilt angle of the rotation axis in the region of 10 - 35° when the measured dimensional deviation is minimal. The CT scanning method also changed from panning to spiral scanning (spiral CT scan) (Jacobson et al., 2000).

In addition, improving the quality of CT images is one of the more current research efforts to facilitate the applications of CT. Houston et al. (2013b) pointed out that the sharpness of the image is crucial for determining the subsequent spatial segmentation of the pore space, which requires a higher resolution at a cost of additional noise. The intensity of X-rays directly affects the signal-to-noise ratio (SNR) and is critical to eliminate the effects of X-ray scattering on the quality of CT images (Gao et al., 2021a; Zhang et al., 2021b). Although conventional image reconstruction methods can obtain high quality images, they are limited by the reconstruction algorithms (e.g., the filtered back projection-FBP algorithm), whose assumptions (e.g., single X-ray photon energy and uniform photon propagation direction) are difficult to meet in reality. Deviations from basic assumptions in the acquisition of CT images would reduce the accuracy of subsequent segmentation and identification of image objects (Zhu et al., 2019)). Recently, deep-learning reconstruction algorithm (DLR) (Franck et al., 2021; Higaki et al., 2020; Steuwe et al., 2020; Urikura et al., 2021) was employed to correct the noise attributes of CT images. A total variable (SART-TV) (Wang et al., 2021b) algorithm-based image reconstruction can better suppress the artifact effect and fixed the problem of sinogram loss for dual-energy CT (DECT) (Kawahara et al., 2021). Zhang et al. (2021d) have alleviated the nuisance caused by the non-ideal focal size of the X-ray source for CT imaging and provided high quality images even under non-ideal X-ray source conditions. Alternatively, progressively shorter scan times (e.g., high-speed CT scans (Kureta and Iikura, 2009; Yan et al., 2021) could avoid motion artifacts (Hegazy et al., 2021; Hsu et al., 2007). The scanning post hoc image summation method (fused CT) reduces image noise (Kobayashi et al., 2020), while the addition of contrast agents (Aslan et al., 2020; Inose et al., 2021) enhances the contrast and image quality of the study area.

High-resolution chromatographic imaging instruments can be used to acquire 3D images of porous materials (e.g., soil samples) and to reveal its structure. It should be noted that quality of the CT image should be priority. However, the image itself only is not sufficient for quantitative analysis, and image processing methods are needed to extract and interpret useful information. It is still challenging for image processing methods to specify hole boundaries or analyze errors, distortions caused by CT reconstruction (Ojeda-Magaña et al., 2014; Zhao et al., 2019).

2.3. Basics of CT image processing

The main image processing tasks in order of processing include: noise reduction, edge enhancement, image segmentation, post-processing, and structure analysis (Schlüter et al., 2014). The most difficult and widely studied part of X-ray CT images is commonly based on thresholding and segmentation processes. X-ray CT analysis is widely related to the thresholding step which is mainly attributed to the calculation accuracy of soil porosity that is linked to the gray value. Gackiewicz et al. (2019) found that the range of the error which is related to the thresholding step was between 15 % and 40 % at the calculation of visible total soil porosity and saturated hydraulic conductivity. Thresholding of the images must be completed prior to analysis according to different histogram shapes, spatial clustering, entropy, object attributes, spatial correlation, and local grayscale surface information (Sezgin and Sankur, 2004). These six categories are used to distinguish between particles and pores and to segment the grayscale map into a binary image to smooth the particle and pore boundaries (Tang et al., 2020). Various statistical and mathematical methods can be applied during this process.

The widely utilized segmentation process mainly includes local segmentation (Schluter et al., 2014) and global segmentation (Lavrukhin et al., 2021) that are based on threshold processing. They transfer images into discrete datasets in order to separate between the solid phase and soil porosity for the quantitative description of porous media. The advanced segmentation techniques are mainly based on local segmentation methods, because the global segmentation method shows less capability dealing with local heterogeneity and is low in global robustness (Iassonov et al., 2009). Houston et al. (2013b) tested four threshold-based segmentation methods (e.g., Thresholding method by Schlüter et al., 2010, Adaptive window Indicator Kriging by Houston et al., 2013a, Fully Automated Extension method based on Otsu by Hapca et al., 2013 and Global segmentation method by Otsu, 1979) commonly used in soil research. The performance of the first three methods are stable in the image process but with different noise levels, the first two methods tend to eliminate small holes and smoothen the surface of larger holes (Houston et al., 2013a). Wang et al. (2021a) proposed that the pore structure of soil samples usually cannot be reconstructed using a single threshold, and it is important to select the appropriate thresholds for each CT image in order to ensure segmentation accuracy. Other commonly used alternatives of segmentation methods include region growing (Gerke and Karsanina, 2021), watershed algorithms (Sun et al., 2019), unsupervised learning and supervised learning (Chauhan et al., 2016) based on clustering, etc. Supervised machine learning (e.g., Neural network, Random forest, Support vector machine, Linear and logistics regression, and Classification trees) that are trained the with well labelled data to better predict outcomes of new data/images are widely adopted (Lai and Chen, 2019; Sidorenko et al., 2021; Wang et al., 2022). However, it requires time, efforts, and a team of data scientists with expertise and experience in specific subject to build, scale, and deploy accurate supervised machine learning model (Hashemzadeh and Adlpour Azar, 2019; Piccoli et al., 2019). Such requirements are especially challenging for image segmentation of naturally heterogeneous materials such as soils and plants. In contrast, unsupervised learning techniques (e.g., Cluster algorithms, Hierarchical clustering, and K-means) based segmentation allow the model to work independently to discover information of unknown pattern or useful features for categorization without manual intervention, more importantly they can perform more complex processing tasks and provide results for users in real time (Hapca et al., 2013; Jafari-Marandi, 2021; Schlüter et al., 2010; Zhang et al., 2008), although the unsupervised machine learning methods could be more unpredictable compared to reinforcement learning, natural learning, and deep learning. which will significantly improve the efficiency. The QCuts-3D that is based on unsupervised segmentation method (i.e., spectral clustering) for automatic segmentation makes it an attractive candidate for 3D aperture segmentation (Malik et al., 2022). In addition, little difference is found in X-ray attenuation between water and organic matter and plant roots (Pot et al., 2015), and it is challenging to distinguish between components of similar densities if X-imaging is not enhanced (Liu et al., 2012; Taina et al., 2008). The use of synergistic unsupervised and supervised machine learning for image segmentation may be a promising solution for this (Alrfou et al., 2022).

Based on the X-ray CT images, Houston et al. (2017) summarized the currently available software tools and methods for evaluating the pore size distribution (PSD). They also evaluated the performance of seven commonly used methods (i.e., Avizo, CTAnalyser, BoneJ, Quantim4, DTM, DFS-FIJI, and 3DMA) on CT images of different soil types and the results showed that PSD as pore volumes per class size derived from the first five methods agreed well with the analytical solution, while the last two methods produced unstable results for complex soil samples. Lucas et al. (2021) developed a new method of Eulerian number X joint Γ-curves to investigate the pore connectivity of pore sizes of more than three orders in magnitude. Information on pore distribution, fractal dimension and pore connectivity could be obtained from images through quantitative image analysis tools such as AVIZO, VGStudio-MAX, and Image J, or even plug-ins for semi-automated image processing like SoilJ (Koestel, 2018). However, Zhao et al. (2019) argued that many already developed integrated software (e.g., Image-Pro Plus, Photoshop, Image J and ArcView) and self-compiling algorithms (e.g., adaptive thresholding algorithm and fixed thresholding algorithm) have difficulty in identifying irregular pores and fuzzy boundaries, so they recommend a clustering algorithm called Adaptive Fuzzy C-mean Method (AFCM) with fuzzy decision making capability. Weller et al. (2021) established an open soil structure database for image analysis and modeling of soil structure, pores and other data. Such work is far-reaching and substantially helpful to scientific researchers, but unfortunately it requires users to upload images that have been segmented. Here, we also provide practitioners with a reference for imaging processes and the selection of segmentation processing methods.

An example of the imaging workflow to demonstrate the procedure for spatial detection and quantification of major pores, solids and gravels in 2D CT soil images, as well as 3D reconstruction of binary images is shown in Fig. 1. Table 1 collated some of the commonly used CT instruments and associated processing methods for soil and plant related analysis.

3. Bibliometric analysis

The Web of Science Core Collection (WoSCC) Science Index Extension (SCI-EXPANDED) contains high-quality publications for research access since 1992. The literature dataset from January 1992 to December 2020 was downloaded from the WoSCC on May 21, 2021 for scientometric analysis. Two query sets were used to for mapping the CT applications in environmental soil and plant related studies (query sets 1) and in all disciplines (query sets 2), respectively. Only articles, notes, letters, books/book chapters, data papers, database reviews, and reviews written in English were retained. The results of the query sets 1 were imported into Endnote for manual filtering following the criteria presented in Fig. S-1. The search resulted in a total of 4,182 publications and only 1,770 relevant studies were included for analysis after screening.

Bibliometric software tools (e.g., VOSviewer, CiteSpace, Histcite, Gephi, Leximancer, etc.) allow the analysis of large amounts of literature data by means of both performance analysis and science mapping. They enable the presentation of the research state and emerging trends of a specific research topic or field (Feng et al., 2021; He et al., 2020; Zhang et al., 2021a). In this study, VOSviewer (v1.6.15 (Van Eck and Waltman,

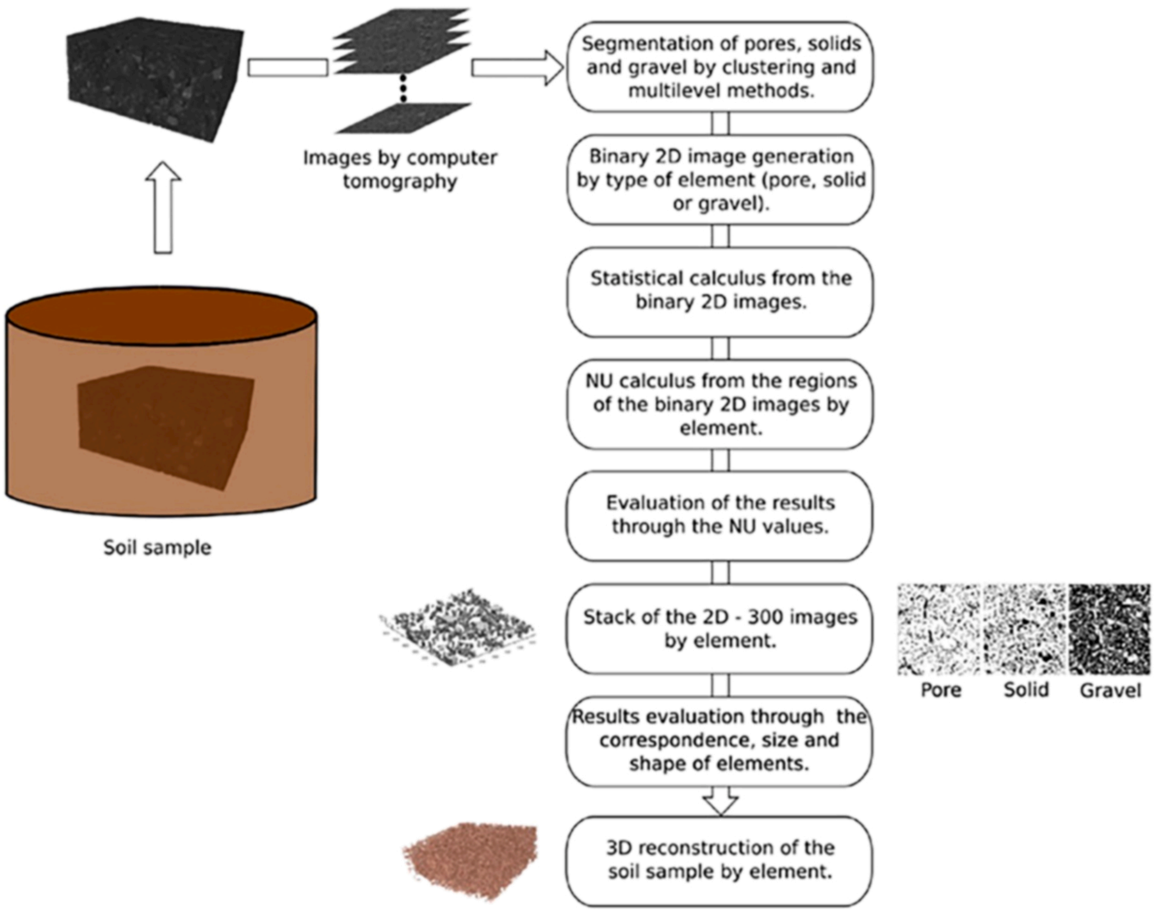


Fig. 1. A case of an imaging workflow: spatial detection and quantification procedures and 3D reconstruction of major pores, solids and gravels in 2D CT soil images, after (Ojeda-Magana et al., 2020). In this case, image segmentation was performed using standard clustering algorithms (k-Means, Fuzzy c-Means) and multilevel thresholding methods (Otsu method), and the quality of image segmentation was evaluated based on NU values (a measure of regional nonuniformity based on unsupervised methods). The closer the NU value is to 0, the better the segmentation is.

2010), from Leiden University, the Netherlands) was used to carry out cluster analysis on keywords in literature. Algorithm of VOSviewer is based on association strength and is preferable to other bibliometric software tools (Lan et al., 2022). It can effectively avoid label overlap and directly reflect the weight of the analyzed data and the relationship between the analyzed data (Fu Jian, 2019).

3.1. *Annual publication trend*

The results of query sets 2 (252,825 studies) show almost the same trend as the query sets 1, but with a 100-fold magnitude (Fig. 2). This means that the application of CT has been in a booming state.

The results of Query sets 1 were divided into four groups that are closely related to plant or soil. The green and brown colors indicate studies directly applying CT visualization techniques in plant and soil science only, respectively (Fig. 2). The number of publications of related studies was small until 2005, with less than 10-article each year. The number of publications started to increase since 2005 with fluctuations, indicating that the application of CT in plant and soil sciences has become increasingly popular. The blue bars started from 2002 refer to publications using CT in geotechnical applications. The purple bars refer to the CT related research work on interactions between plant roots and soil, which started to grew slowly since its occurrence in 2006.

3.2. *Co-occurrence analysis of keywords*

A total of 652 keywords (filtered by keyword frequency ≥5 times) were selected for research related to CT in soil and plant science, with the most frequent keywords being "computed-tomography", "agriculture", "engineering", "materials science", "geology", "plant sciences", "soil", "environmental sciences & ecology", "quantification", "forestry". Fig. 3 shows a total of nine clusters of different colors, each one representing a corresponding research theme. The larger clusters are yellow, blue, green, and pink, which represent the different directions of CT applications in soil and plant science, respectively. For instance, the yellow cluster focuses on agricultural soils and ecological issues, and includes "soil structure", "solute transport", "organic-matter", "water resources and water-flow". The pink cluster concentrates on research on geotechnical engineering. The green and blue clusters are both about plant science, but the green focuses more on forestry and material properties of wood, while the blue focuses more on using CT tools to show plant growth and root systems, as well as water use.

3.3. *Burst of keywords*

A keyword with strong emergence can reflect a new perspective with a high impact in a certain period of time, thus showing a phase of academic frontier. The word frequency exploration was performed in CiteSpace, and the top 15 keywords related to CT scans in soil and plant scientific research from 2010 to 2020 were used (Fig. 4). From

Table 1
Selected computed tomography (CT) instruments and processing, modeling methods for soil and plant analysis.

Author	CT model/parameters	Resolution	Image segmentation processing	Other information
Zuniga et al. (2019)	Phoenix Nanotom X-ray μCT; 65–82 kV, 167–370 μA	Resolution of pores > 60 μm (2 voxels) in diameter	Select two thresholds by applying Otsu algorithm twice	Global segmentation technique, is a robust method when applied to the image with low complexity
Budhathoki et al. (2022)	Medical X-ray CT; 140 kV, 140 Ma	Voxel size of 0.35 * 0.35 * 0.625 mm^3	Adaptive local threshold method	Used for processing low contrast images; can solve the problem of uneven staining
Pot et al. (2020)	HMX CT; 225 kV	Spatial resolution is 24 μm and 54 μm	Otsu's global method	Thresholding method based on clustering; the small pores are segmented into continuous holes at high initial resolution; not sensitive to noise reduction level
			Hapca's method	Segmentation results are the most robust; porosity is less at the point of minimum contraction; when the voxel strength is close to the solid phase of liquid or oil phase, the performance is poor
			Schlüter's method	Small pores are divided into independent clusters at high initial resolution; penetration holes may break at high noise reduction levels
			Adaptive-window Indicator Kriging method of Houston	Small pores are divided into independent clusters at high initial resolution; when the solid phase dominates, porosity may be masked
Martín-Sotoca et al. (2018)	Metris X-Tek X-ray micro-CT system; 160 kV, 201 μA	Resolutions of 30 μm (voxel length size)	The maximum entropy method	Thresholding method based on entropy information of gray distribution; doesn't particularly rely on bimodal histograms; underestimated the true total number of pores
			Singularity-CV segmentation method	High sensitivity; some small pores may be misidentified; underestimated the true total number of pores
Cheng and Wang (2020)	The BL13W beam line of the Shanghai Synchrotron Radiation Facility	–	Improved region growing segmentation method	The interface of the lowest (P1) and highest (P3) phases is effectively avoided to be misidentified as the P2 phase; more time-consuming; cannot be used for objects with more than three phases
Lavrukhin et al. (2021)	SkyScan-1172 XCT desktop scanner	Resolutions of 15.85–15.88 μm	Convolutional neural network + local segmentation	Independent, automatic, high throughput; prediction quality depends on the completeness of the real sample used for training
Malik et al. (2022)	–	–	Unsupervised QCuttS-3D spectral clustering segmentation method	Unsupervised, automatic, class-independent significant object detection method; area boundaries can be better preserved; identification of changes in particle geometry is more robust
Sun et al. (2019)	EinScan-SP 3D laser scanner	Scanning accuracy is 0.05 mm	Improved watershed algorithm	Suitable for separating particles in contact; overcome the oversegmentation: not available for complex soils
Chirol et al. (2021)	Nikon Metrology XT H 225 X-ray μCT; 205 kV, 46 μA	The voxel grid is 62.5 μm	Hysteresis thresholding method + Frangi filter	Can distinguish living roots from dead tissue; live roots attached outside the core are mistaken for necrosis; parameters need to be manually selected in advance
Maenhout et al. (2019)	'HECTOR' X-ray μCT scanner; 150 kV, 50 W; 0.5 mm Cu filter	2000x 2000 voxels	New division protocol	Based on the protocol proposed by Pfeifer et al. (2015); root volume segmentation was less affected by soil type; can be applied to complex root systems; time-consuming and requires proactive intervention
			RooTrak	Free software packages; not suitable for large sample systems such as mature plant roots
			Region growing algorithm for VGSTUDIO MAX 3.2	Calculate the tolerance value based on gray level; the scope of application is limited to small and uniform soil cores; tolerance values need to be chosen carefully

2010–2015, research perspectives were mostly focused on the three-dimensional quantification of soil or plant health indicators by X-ray CT, including the aggregate structure of the soil and the spatial heterogeneity of the voids around the structural units, water flow and biology of the soil, and plant growth. From 2015–2020, a large number of qualitative and quantitative research cases on saturated hydraulic conductivity and preferential water flow emerged, indicating that the research area is more refined in terms of the hydraulic properties of soil. Particular attention is paid to the interconnection and comparison of values obtained from X-ray CT measurements with model parameters, which contributes to the accurate estimation of hydrodynamic behavior of soil.

4. CT applications in soil science

Improvements in technology and equipment, as well as cost reductions, have made CT available in soil science (Torre et al., 2020). Petrovic et al. (1982) pioneered the application of CT in soil science and demonstrated a linear relationship between soil capacity and X-ray attenuation, opening up a grand opportunity to use CT in soil science. CT provides useful information to better understand and describe the porous media structure, hydraulic properties, heat transfer, carbonate dissolution, solute transport, gas exchange, rheological properties, root development, and biological activity of soils while preserving the integrity of the sample.

4.1. Soil physicochemical properties quantification using CT

4.1.1. Soil pore space and pore network structure quantification

In terms of pores, soil structure can be defined as "the combination of different types of pores and aggregates" (Pagliai and Vignozzi, 2002). Qualitative and quantitative soil pore structure is crucial for soil research, because the spatial arrangement and pore connectivity of soil pores control the soil-plant-atmosphere interface (Dexter, 1988). The difference in X-ray attenuation between soil pores and soil solid components is significant and contrasting (Young and Crawford, 2004), which make it relatively easy to distinguish pores in the CT reconstructed images. Therefore, X-ray CT is widely used to describe the shape, size, connectivity, arrangement, volume, distribution, geometry, and orientation of pores to calculate soil porosity (Hu et al., 2020; Peyton et al., 1992), as well as to scan the three-dimensional microstructure of soil aggregates (Alskaf et al., 2021; Pagenkemper et al., 2015). The porosity of the soil (e.g., total porosity, microporosity, macroporosity) can also be obtained from the gamma ray attenuation

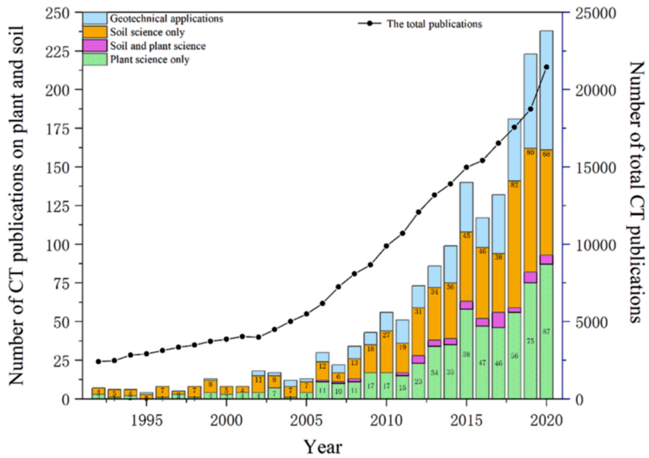


Fig. 2. Annual analysis of publications on application and research of computer tomography (CT) technology in soil and plants science with query sets 1 (pillar, with a total of 1,770 publications) and all applications and research related to computed tomography techniques with query sets 2 (line, with a total of 252,825 publications) based on the SCI-EXPANDED, SSCI, CCR-EXPANDED and IC databases of the Web of Science Core Collection (WoSCC) between 1992 and 2020. The query sets 1 used for the literature search are: TS= ("gamma-ray CT" OR "X-ray CT" OR "computed tomography" OR "micro-CT") AND TS= ("soil" OR "ped*" OR "ground" OR "plant" OR "wood" OR "flora" OR "botany" OR "vegeta*" OR "grain" OR "food" OR "timber" OR "root") NOT TS= ("medicine" OR "acromegalic" OR "patient" OR "operation" OR "human" OR "disease" OR "cancer" OR "coronavirus" OR "radiologists" OR "cardiac" OR "COVID" OR "surgery" OR "hepetic" OR "skele*" OR "dentine" OR "teeth" OR "canine" OR "pediatric" OR "cervical" OR "intestines" OR "bubble" OR "bone" OR "chest" OR "pancreas" OR "mandibular" OR "dairy" OR "subgrade" OR "blood vessels"). Query sets 2: TS= ("gamma-ray CT" OR "X-ray CT" OR "computed tomography" OR "micro-CT").

method (GAM), where the bulk density is the only parameter required to evaluate these pores (Pires, 2018). However, it should be noted that even though the advanced X-ray CT techniques have multiple advantages, only a relatively small fraction of pore scape (visible porosity) can be characterized in real soils, as a fraction of the pore space below the scan resolution remains unidentified.

Tillage directly affects porous media structure (Schluter et al., 2020; Zhou et al., 2013), which can be measured by the CT. The X-ray CT results showed that conventional tillage generally increases macroporosities (>500 μm) as well as anisotropy and fractal dimension of pores, but interrupts vertically well-connected bio-pores, shrinkage cracks, and decreases biota (Guo et al., 2020; Hellner et al., 2018). While no-tillage leads to increasing microporosity (100–500 μm) as well as the connectivity of pores (Guo et al., 2020). Bio-tillage is believed to improve soil structure (Elkins, 1985) and more intuitive evidence can be achieved by CT to determine the effect of the resulting bio-pores on the growth of plant roots (Zhang and Peng, 2021). The double logarithmic stress-macroporosity diagrams (σPMP) based on X-ray CT studies indicated that a critical mechanical precompressive stress value appears to exist around 100 Kpa for strip tillage, cover tillage, and no-till, where the macroporosity and connectivity reach the irreducible minimum with considerable increases in anisotropy (Pöhlitz et al., 2018). In addition, examination of soil from a 23-years experimental plots showed long-term manure can change the microstructure of the soil (Yu et al., 2020a), i.e., increasing the soil bulk density and decreasing soil porosity (Zhou et al., 2013). Application of inorganic and organic compound fertilizers tends to decrease soil bulk density and increase connectivity (Fang et al., 2021; Lan et al., 2020). Moreover, rotation diversification was also believed to increase soil organic carbon (SOC) and improve soil pore characteristics (Singh et al., 2020a). Furthermore, biochar-remediated soils would lead to significant pore network reconfiguration (Yu and Lu, 2019), while the distribution of soil pores was optimized in calcium lignosulfonate-modified swelling soils (Wu et al., 2020).

Soil pore network is complex and pore size distribution generally can only be indirectly estimated through soil water retention curve (SWRC) (Singh et al., 2021a). X-ray CT on the other hand provide precise and effective characterization of the number, shape, size, orientation, connectivity, curvature, and spatial distribution of pores, enabling the acquisition of a complete 3D structure. However, the pore size volume estimated by SWRC is usually larger than that calculated by X-ray CT (Shein et al., 2016). In addition, only a small soil sample can be scanned with the high-resolution X-ray CT (Singh et al., 2021b) which still present a method limitation.

CT can be also employed to characterize the soil disturbance state and damage. The mean density (MD) and standard deviation (SD) of X-ray CT images are used to define the soil disturbance perturbation state intensity, where the SD value is considered to be more appropriate (Li et al., 2019). The CT scanning combined with 3D image reconstruction technique could also be used to build soil morphology, which verified freeze-thaw cycles (FTCs) induced damage to soil microstructure (Fan et al., 2021; Gao et al., 2021b; Liu et al., 2021b; Ma et al., 2021) and the small pore expansion changes eventually formed cracks (Xu et al., 2021a). Xu et al. (2019) analyzed the crack morphology, shear strength under salt erosion and FTCs. Their results showed that the fractal dimension implied structural damage and the spiral CT mean value behaved similarly to the variation of shear strength. Multiple FTCs leads to increase total porosity, pore connectivity, total available water, and hydraulic connectivity (Liu et al., 2021a). If the soil was treated with lime, structural damage could be reduced and the resistance of soil to freeze-thaw damages was improved (Nguyen et al., 2019). Mady and Shein (2020) used X-ray CT to monitor the changes in pore space during the wetting and drying cycles (WDCs). They found that the pore structure differed under the two WDCs curves at the same soil water potential, which could potentially explain the reasons of hysteresis in the SWRC. Continuous WDCs increased soil porosity and improved pore connectivity (Pires et al., 2020), and the magnitude of change in macroporosity was greatest in clay-rich soils (Diel et al., 2019). Xu et al. (2020) found that the change of fracture rate in loess is synchronous with CT average value after WDCs. Yu et al. (2020b) suggested that the coefficient of variation associated with X-ray CT slices can reliably evaluate soil heterogeneity.

The conceptual model in conjunction with laboratory measurements is needed to correctly describe and verify the complex soil pore structure and properties (Lamande et al., 2021; Li et al., 2018). The loess soil structure index (LSI) could be expressed as a nonlinear function related to CT image parameters (Tang et al., 2021). Complex quantification of soil pore networks can also be achieved by multifractal model analysis, the gray value (GV) synthetic maps created with the truncated multifractal method (TMM) are able to resemble real CT soil images (Torre et al., 2020). The multifractal parameter is significantly correlated to the macropore number and was the main variable controlling the expression of the total porosity in the model (Ju et al., 2021). While Perrier et al. (2021) discussed from several perspectives of mathematical modeling, physical modeling, computer modeling, and statistical modeling, they concluded that multifractal modeling for spatial distribution of soil pore and solid mass is not applicable. Tang et al. (2020) developed MATLAB-program based soil microstructure analysis system (SMAS), which can effectively identify soil microstructures and extract information accurately.

The accuracy of the description of the soil microstructure largely depends on the resolution of the CT instrument. However, Baveye et al. (2018) stressed that a portion of the soil pore space was still difficult to identify even CT resolution could be increased to a fraction of a micron (i.e., sub-resolution pores smaller than the resolution of the CT image). An innovative solution for this issue is to concurrently scan soil samples with three distinct X-ray energies and specific stains (Baveye et al., 2017), which was then tested by Fonseca et al. (2019) who applied X-rays and γ-rays CTs to scan the same sample with a double ratio

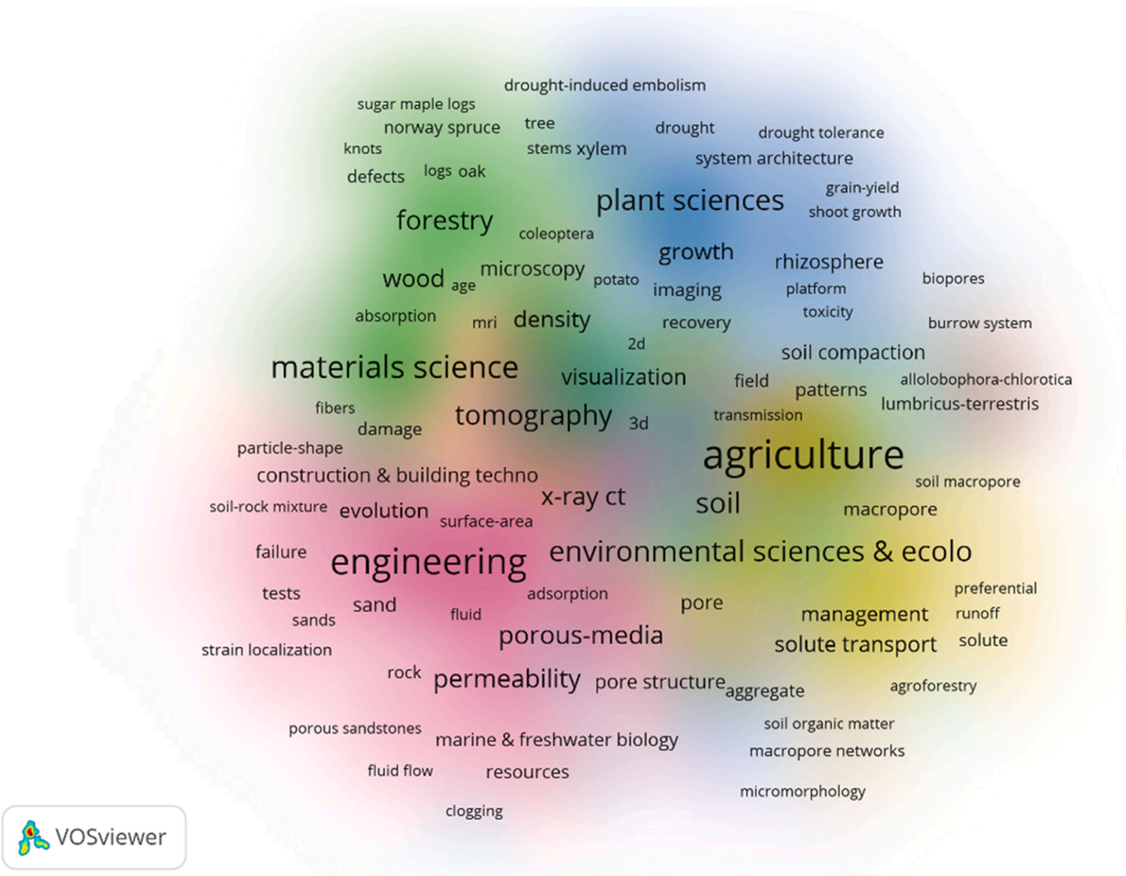
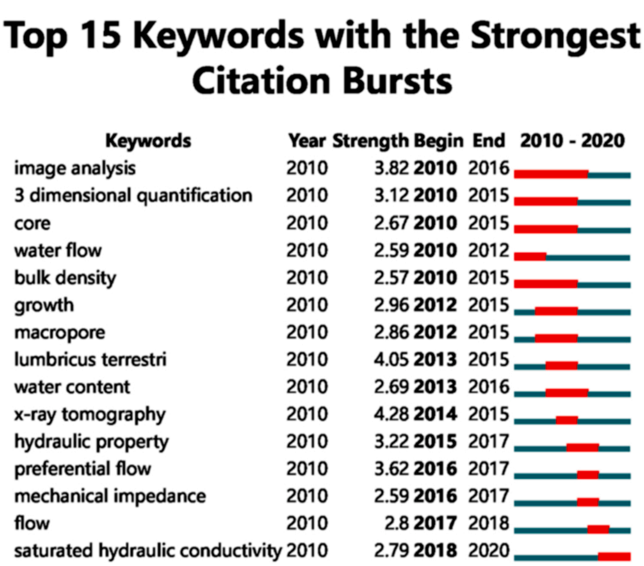


Fig. 3. Keywords co-occurrence network visualization is generated by VOSviewer. Each color represents a topic cluster, where the font size and density (background color) of the keyword indicate total link strength (TLS). A greater font indicates greater TLS, and the closer the distance between keywords, the higher the relevance of these studied topics.



Top 15 Keywords with the Strongest Citation Bursts

Keywords	Year	Strength	Begin	End	2010 - 2020
image analysis	2010	3.82	2010	2016	
3 dimensional quantification	2010	3.12	2010	2015	
core	2010	2.67	2010	2015	
water flow	2010	2.59	2010	2012	
bulk density	2010	2.57	2010	2015	
growth	2010	2.96	2012	2015	
macropore	2010	2.86	2012	2015	
lumbricus terrestri	2010	4.05	2013	2015	
water content	2010	2.69	2013	2016	
x-ray tomography	2010	4.28	2014	2015	
hydraulic property	2010	3.22	2015	2017	
preferential flow	2010	3.62	2016	2017	
mechanical impedance	2010	2.59	2016	2017	
flow	2010	2.8	2017	2018	
saturated hydraulic conductivity	2010	2.79	2018	2020	

Fig. 4. Information about keywords with the strongest citation bursts from 2010 to 2020 by CiteSpace.

method to evaluate the ray energy levels and mass attenuation. The results indicated a maximum error of only 1.5 % compared to the NISTXCOM database, which is promising for high-precision characterization of soil structure. The alternative is the use of ^{1}H Nuclear Magnetic Resonance (NMR) relaxation to assess the pore size distribution in soil samples over a wider range, from nm to mm, but at a cost of time compared to X-ray CT scanning (Jaeger et al., 2009; Pohlmeier et al., 2018). It is time consuming and laborious for the focused ion beam plus scanning electron microscopy (FIB-SEM) that characterize the structure of soils at the nanoscale level (Gerke et al., 2021).

4.1.2. Soil water and air quantification

Soil water and air movement are usually related to the pore size distribution, as well as the connectivity and tortuous degree of pores (Osozawa, 1998). In general, a large diameter and well-connected pore network usually facilitates water penetration and gas aeration (Rabot et al., 2018). As early as 1992, CT imaging was used to measure soil water distribution (Hopmans et al., 1992) and indirectly infer soil hydraulic functions. Kido and Higo (2020) studied the connectivity between water and pores using X-ray μCT and related image processing. They evaluated the morphology of soil pore full of water, liquid bridges, and water-gas interfaces, the hydraulic conductivity is related to the uniformity of porosity, pore size distribution, and pore connectivity (Wang et al., 2021a). Mady et al. (2021) demonstrated that a greater influence of mesoporosity than macroporosity on saturated hydraulic conductivity values (K_s) of undisturbed soil core samples from four depths (i.e., 5–10, 15–20, 25–30, and 40–45 cm) using X-ray CT. More mesoporosity results in greater K_s, while more closed pores and poorer poor connectivity lead to smaller K_s. The pedotransfer functions (PTFs) used to predict K_s proposed by Koestel et al. (2018) is based on three parameters (i.e., the pore geometry factor, the critical pore diameter, and the effective porosity), the last two of parameters determining variation in K_s were calculated from X-ray CT measurements. Smet et al. (2018) reported that the usage of pore size distribution estimated by

X-ray CT images increased accurate estimation of SWRC and hydrodynamic behavior. In fact, the transport of water and air is limited by the soil layer with the smallest macroporosity (Katuwal et al., 2015b). In the case of low water potential, Zhang et al. (2019) stated that the estimation of K_s and air permeability are more correlated with the mean diameter of the limiting layer (MDLL, the layer with the lowest value of macroporosity) than other pore characteristics (e.g., total porosity, connectivity, etc.) derived from X-ray CT. Schluter et al. (2020) restructured pore structures with CT images for Stokes-Brinkmann solvers to model saturated and near-saturated water flows under different tillage treatments. The resulted K_s was greater than estimated K_s with Hood Infiltrometer and Tension Disc Infiltrometer and best correlated was established with pore space attributes.

The sensitivity of magnetic resonance imaging (MRI) to ^1H makes it ideal for studying aqueous liquids (Carlson, 2006), but it can cause distortions in the generated MRI images if the porous medium has an inherent relaxation time (Sadeghi et al., 2017). Another commonly used non-invasive technique, Electrical Resistance Tomography (ERT), uses the electrical conductivity of water to distinguish between the effects of capillary barriers, water movement, and steam intrusion in complex geological environments (Daily et al., 1992; Ramirez et al., 1993). Cimpoiasu et al. (2020) compared ERT and X-ray CT and developed a new relationship for the data conversion between them, which could be conducive to our further exploration of soil hydraulic function. However, it is noteworthy that CT is still not flexible enough to realize on-site delayed monitoring (especially for the movement of water vapor in pores) (Liu et al., 2022). In addition, CT is not suitable for soils where most of the pores are finer than the CT resolution, because CT images of such soils do not reveal any of the underlying architecture, pore geometry, or pore connectivity.

Entrapped air in near-saturated and saturated soils, mainly in the form of single pore bubbles, clusters or ganglia, may clog the pores and thus affect water transport (Faybishenko, 1995). CT scanning combined with the priming and infiltration tests were used to obtain information on the spatial distribution of bubbles, together with image processing to obtain the morphological information on the size, shape and thickness of bubbles (Muller et al., 2019a; Princ et al., 2020). If the morphology and transformation of pore air and pore water in the two phases of drying and wetting were specified by scanning at different water retention states, this would be useful to elucidate the reasons for the lag in the soil water characteristic profile exhibited after drying or wetting (Kido et al., 2020). However, it should be noted that X-ray CT analysis is often based on local tomography, multiresolution analyses are required to obtain a global view of the whole sample.

4.1.3. Soil organic matter quantification

Combination of CT with other methods (staining or labeling) can be used to facilitate visualization of soil organic matter (SOM). Because the X-ray attenuation coefficient of SOM lies between the pore space and the soil matrix, and the SOM is closely related to the mineral surface, it is difficult to separate it in the image using CT detection alone (De Gryze et al., 2006). For example, CT can be combined with staining technology to identify SOM (Chenu and Plante, 2006). Osmium tetroxide (OsO_4) vapor can react with unsaturated carbon bonds in organic compounds with a clear absorption edge at a specific photon energy. OsO_4 was used as a dye to locate the distribution of SOM in soil aggregates by Peth et al. (2014). It is noted that OsO_4 cannot dye all types of organic materials evenly (Zheng et al., 2020) and volatilized OsO_4 is also toxic to human beings. Maenhout et al. (2021) tested four alternative chemical stains, i.e., lead nitrate ($Pb(NO_3)_2$), lead acetate (PbAc), phosphomolybdenic acid (PMA) and silver nitrate ($AgNO_3$), to enhance the contrast of images in soils containing specific minerals. Their results indicated that only PMA was comparable in staining ability to OsO_4, the other three stains can only perform well in sandy soils. However, PMA infiltrated soils suffered from only a small increase in X-ray attenuation when SOM was subjected to polychromatic X-ray beams. Liquid-phase osmium-thiocarbohydrazide-osmium (OTO) was also used as alternative for staining to reduce the risk OsO_4 (Arai et al., 2019). In addition, liquid and gaseous iodine (I_2) and heavy elements (e.g., Ag, Cu, Co, Fe, Mn, Mo and Zn) are often used for staining, with Ag performed best among the heavy elements as other heavy elements are naturally present in the soil. Under K-edge SR-μCT scanning, gaseous I_2 was treated more effectively than Ag (Lammel et al., 2019).

In addition, samples for CT scan are usually prepared with resin embedding techniques that immobilize the sample and better preserve the sample microstructure. However, all resins contain carbon that influence mapping organic carbon concentration distributions (Koestel et al., 2022). Moreover, organics with more double bonds lead to stronger signals, which requires to systematically study the correlation of osmium signal strength with the organic type (Peth et al., 2014). It is also not possible to stain the fraction of SOM that cannot be represented by unsaturated carbon as well as the fraction of SOM that OsO_4 cannot diffuse (gas phase) and permeate into soil matrix. For SOMs with sizes below the effective resolution of CT, it is difficult to detect the absorbed stain in order to provide sufficient contrast for SEM/EDS and μCT analyses (Arai et al., 2019). SR-μCT has the best resolution of μCT to detect SOM (Lammel et al., 2019), but its high expense limited its broader application. Overall, stains have the potential to identify discrete SOMs in soil, while staining make it more difficult to segment CT images.

Alternatively, it is also possible to quantify SOM without staining. Neutron CT and X-ray CT imaging have received attention due to their complementary attenuation properties. Koestel et al. (2022) introduced time-of-flight (TOF) neutron imaging into soil research, and combined it with X-ray CT imaging to map soil SOM concentration and minerals. However, their findings argue that this combination is not suitable for distinguishing SOM and clay minerals at the voxel scale, because it was impossible to differ SOM from clay minerals, especially when these components were mixed together. In addition, the low SNR of current TOF neutron imaging poses a challenge for image analysis.

4.1.4. Other soil properties quantification

CT can also be used to better understand soil heat transfer, which is largely determined by soil water content, bulk density, soil mineralogy, and porosity etc (He et al., 2018; Liu et al., 2021c; Zhao et al., 2022). During the drying process, X-ray CT estimated total porosity (and mesopore content) were negatively correlated with the soil thermal diffusivity and conductivity as well as volumetric heat capacity (Mady et al., 2020). In addition, Mady et al. (2020) reported that the influence of visible total porosity is larger than pore size distribution (mesopores content) on the values of soil thermal diffusivity and other soil thermal parameters (soil thermal conductivity and volumetric heat capacity).

In addition, CT images have been used to determine the effective cation exchange capacity (CEC) of soil cores (Keck et al., 2017) and soil colloids (Scotson et al., 2019). Because of the high adsorption affinity between Br^{2+} and cation adsorption sites (CAS), Keck et al. (2017) used Br^{2+} as a contrast agent to visualize CAS bound by Br^{2+} in 3D images reconstructed by X-ray CT by differential imaging technique, which was used as an indicator to measure CEC. However, this method also suffers from estimation errors in some cases, for example, the CEC of organic matter-rich samples is overestimated and the CEC of clay-rich samples is underestimated. Leue et al. (2020) combined the image data measured by X-ray CT with OC content and CEC values for specific types of coatings on the surface of soil core macropores for the purpose of quantitative estimation. They found that OC content and CEC accounted for about one-third of all macroporous surfaces, and biopores dominated the storage of OC. They also noted that larger cores were more representative, but the resolution of the X-ray CT spatial measurements limited the size of the samples, and a regrettable drawback in their study was that the thickness of the macropore surface layer was not very clear and was difficult to distinguish from the soil matrix. The use of other techniques (e.g., diffuse reflectance infrared Fourier transform (DRIFT) spectroscopy to obtain valid CEC information (Leue et al., 2019)), can yield more accurate information in smaller soil slices. To better describe soil processes, the spatial information from these 2D mappings can be interpolated into a 3D image of the soil column. This versatile combination allows correction of geometric distortions in the 3D image caused by the sample during the cutting process and also uses the CT image to help map other soil chemical parameters, pointing to a future connection between the 3D visualization of not only the physical aspects of soils but also their chemical characteristics (Hapca et al., 2011, 2015; Schlüter et al., 2019). Interestingly, researches showed that combined TOF neutron imaging and X-ray CT imaging is useful for the identification and characterization of substances such as plastic beads that are larger than the pixel size of the detector (Koestel et al., 2022), and the microplastics in the millimeter size range (Tötzke et al., 2021).

4.2. *Soil biology quantification*

Soil microbes (Juyal et al., 2019; Kravchenko et al., 2019; Portell et al., 2018) and macrofauna (e.g., ants, termites and earthworms (Cheik et al., 2019) as well as their activity dynamics (Ray et al., 2019) and burrow distribution (Schomburg et al., 2019; Zhang et al., 2018) are required to understand soil biochemical processes, but the complex heterogeneity and opacity of the soil environment makes such study difficult to be conducted. Compared to the 2D methods, 3D techniques (e.g., X-ray CT) can describe the natural soil pore space-natural habitat more accurately (Balseiro-Romero et al., 2020). The behavior of earthworms in the soil can be demonstrated by particle tracer in combination with X-ray CT (Capowiez et al., 2021). For example, Guo et al. (2021) has shown that the earthworm activity increases soil anisotropy and reduces pore connectivity, possibly because some earthworm species (e.g. *Lumbricus terrestris* (Jegou et al., 1999) prefer to use burrows as permanent structures. For microorganisms, soil aggregates affect significant soil microbial activity and distribution (Liang et al., 2019). Harvey et al. (2020b) suggested that microbial communities may be able to be protected by the complex porous structure of aggregated soils from specific environmental disturbances (e.g., heat stress). Ruiz et al. (2020) studied the effects of soil nitrogen transport and concentration distribution on microorganisms based on the obtained image model and the respiration experiment of microorganisms. The use of CT to monitor microorganisms in soil habitats was described in detail by Harvey et al. (2020a). Unfortunately, the examples given by Harvey et al. (2020a) only used CT to examine the habitat structure of microorganisms, and the inspection of microorganisms often combined with methods such as microbial respiration and abundance, and DNA extraction. Limited by CT resolution, there is currently no way to directly visualize bacteria and archaea in soil only by CT. Although in situ CT detection of fungi is hampered in some ways (e.g., high electron density of soils, and the same attenuation of X-ray absorption between fungi and water (Falconer et al., 2012)), synchrotron X-ray computed tomography (SXRCT) revealed significant differences in hyphal morphology (Keyes et al., 2022). However, the findings could not represent the hyphal population of the entire soil, as their method could only extract hyphal morphologies in soil pore space. Visualization techniques like Fluorescence staining stated in Section 4.1.3 to strengthen the contrast of target objects with background (Juyal et al., 2021), and the integration of multiple techniques, as Schlüter et al. (2019) discussed, may open up new avenues for research.

It is worth noting, however, that ionising radiation can be harmful to organisms and in severe cases can lead to damage to imaging features (Odstrcil et al., 2019). Therefore, it is necessary to consider the problem of radiation dose without affecting imaging (Harvey et al., 2020a). In general, as mentioned by Vogel et al. (2022), interdisciplinary research combining soil structure and soil ecology is a possible way in the future, which will help to understand the spatial development of soil organisms and realize the soil functions better, such as carbon sequestration and turnover, the degradation of pollutants.

4.3. *Using CT to quantify fluid migration in soil*

CT has also been used to monitor fluid transport in soils. In general, it provides the detection of the spatial distribution of chemicals after their application to the soil.

4.3.1. *Preferential water flow*

Preferential water flow in soils is a widespread phenomenon, classified as finger, funnel, and macropores (Lissy et al., 2020). Current research on preferential flow focuses on two extreme scales (pore scale and larger management scales such as fields and watersheds) and the continuum between them (Jarvis et al., 2016). The dye tracer technique is usually used as an effective and commonly used method to evaluate preferential flow in soil due to its low toxicity, high visibility, high mobility, and no accumulation in soil plants and animals (Sander and Gerke, 2007; Yi et al., 2020), but it may be adsorbed to the macropores walls and result in inaccurate and time-consuming estimates (Jarvis et al., 2016). In contrast, X-ray CT can accurately quantify soil pore space above the resolution and it's possible to do dynamic experiments with smaller samples in a synchrotron that have been recognized worldwide. The combination of staining and CT imaging can monitor the flow paths and preferential flows in soil porous systems (Sammartino et al., 2015; Zhang et al., 2016), providing us with the possibility of in-depth understanding and analysis of flow mechanisms associated with soil structure, from pore to pedon scales. However, the field of view and image resolution are paradoxes (Jarvis et al., 2016) and there are changes in matrix conductivity between samples, which make it difficult to account for variability in tracer migration and mobility at low resolution (Katuwal et al., 2015a).

The combination of CT with numerical simulation has seen growing trend in soil science. However, the success of numerical model simulations depends mainly on the values of its parameters. Can the structural data obtained from CT images replace the parameters of the water flow model? Lissy et al. (2020) pointed out that the kinematic wave (KW) model overestimated the preferential flow in large pores, because the parameters of the dual porosity model need to be further related to the actual sample structural properties. In general, the parameters of these models are obtained by inverse calculation and deduction of fundamental properties, but the operability of experiments and the complexity of reasoning limit their applications. They coupled the improved kinematic dispersive wave model (KDW) (Di Pietro et al., 2003) with the matrix flow equation to compare the two versions (classical: parameters obtained by inversion from calibration data; advanced: parameters evaluated from CT images) by experimental observations. In their experiments, the inverse parameters provided a better simulation. In contrast, the advanced version overestimated the water exchange, but had the advantage of reducing the number of parameters that need to be optimized. Li et al. (2020a) combined X-ray μCT and the Brooks-Corey model (Brooks and Corey, 1965) to predict the effect of capillary suction during water transport, they found that capillary suction is determined by pore size and phase morphology simultaneously, and μCT has satisfactory results in obtaining parameters of air entry pressure and pore size distribution index. For modeling of preferential flow, in addition to providing commonly used pore parameters and permeability parameters, CT also supports the construction of a pore topology network model, which can be used to analyze flow characteristics. After determining the ease of fluid flow in all voids, an algorithm is further used to compare the road resistances of different paths in the topology, and the preferential path of fluid flow can be determined. For the selection of the preferential path, Ant Colony Algorithm, Genetic Algorithm, and Dijkstra Algorithm are the commonly used choices (Meng et al., 2020).

In general, modeling of soil preferential flow still needs to be further investigated, especially in terms of correctly modeling water exchange between the pore space and the soil matrix (Lissy et al., 2020) and quantifying water fluxes, rainfall or groundwater recharge. It is expected that novel CT imaging techniques and processing tools can improve our understanding of preferential and nonlinear transport processes in soils (Tan and Dong, 2022).

4.3.2. Solute transport

The two-dimensional velocity distribution of water and solutes in soil cores can be determined by measuring breakthrough curves (BTCs), which further allows the development of functions for calculating the pore water velocity distribution as a function of soil cores position (Anderson et al., 1992). Nondestructive in situ sampling techniques (micro suction cups (Göttlein et al., 1996) and microdialysis method (Hammarlund-Udenaes, 2017)) are usually required to establish the BTCs of soil solute concentration (Petroselli et al., 2021). Another method to build BTCs is to collect leachate from soil columns (Hochman et al., 2021), which can be combined with convective dispersive equation (CDE) to simulate the spatio-temporal change of solute concentration (Zhen et al., 2017). In addition, the use of X-ray CT to observe soil solute transport processes usually requires the addition of tracers (e.g., contrast agents, adulterants, and heavy elements). Scotson et al. (2021a) employed soluble iodinated contrast agents to simulate field-applied solutes (e.g., soluble fertilizers or pesticides) and X-ray CT was used in combination with modeling to demonstrate the solute leaching process in the field. Guber et al. (2021) showed a new segmentation method based on the combination of mass calculation of dopant mass attenuation coefficients in CT image voxels and mass conservation equations for dopants in soil, which can more realistically describe the spatial distribution of soil liquids, but this can only be applied with dual-energy monochromatic X-ray beams. The use of single-exposure or dual-energy X-ray CT to identify heavy elements in soils was considered to be flawed, because ambiguities in the reconstructed images of the former leads to errors and the latter appearing less reliable at atomic numbers greater than 40 (Nakashima, 2021). To solve this issue, the triple-exposure CT method that can identify different material components with the same or very similar attenuation coefficients is introduced to identify heavy elements (Niu et al., 2020). Nakashima (2021) used a triple-exposure X-ray CT method to nondestructively characterize Pb concentrations in simulated contaminated soils. It is also possible to obtain distribution and migration information of elements or contrast agents by superimposing fluorescent displays (Muller et al., 2019b) onto the CT-obtained 3D pore network (Soto-Gomez et al., 2019). Gattullo et al. (2020) investigated the microscopic and large-scale distribution and morphology of soil chromium (Cr) contamination combining μCT, Micro X-ray fluorescence (μXRF), and field emission scanning electron microscopy coupled with microanalysis (FEGSEM-EDX). Nevertheless, it is noted that tracer addition may make image processing more difficult sometime. The 4D description of 3D-CT image analysis based on time series will be very beneficial to the estimation of motion (Heyndrickx et al., 2018), which requires speeding up the establishment and application of 4D-CT models.

Neutron tomography is a commonly used method to measure the flow and distribution of water (solute) in porous media (Zarebanadkouki et al., 2015). Unlike X-ray CT, which is a function of atomic number and density, the principle of neutron tomography is based on the interaction of neutrons with nuclei, resulting in a strong reaction between neutrons and hydrogen. Therefore, the advantages of neutron CT lie more in the high sensitivity to light elements (e.g., water containing hydrogen, hydrocarbons) and isotopes, the high penetration of metals (e.g., aluminum, titanium, lead), and the much lower radiation damage than X-rays (Tengattini et al., 2021). Some experiments have demonstrated that it can be used to quantify the advance of fluid fronts (Jailin et al., 2018; Tudisco et al., 2019). Although the motion of the fluid caused distortion of the reconstructed volume, the distortion problem might be partially overcome if the advancing velocity was measured at fronts with the same rotational direction (i.e., all fronts had the same type of distortion), or if the data were analyzed as a sequence of projected times rather than as a reconstructed tomographic volume (Tudisco et al., 2019). The key to most flow studies involving neutron tomography is to run them from dry conditions (Deinert et al., 2004; Tengattini et al., 2021). In addition, the scattering effect of neutrons should be considered, otherwise the water content may be underestimated (Tengattini et al., 2021).

5. CT applications in plant science

Nowadays CT is mainly applied in anatomy and analysis of plant morphology and structure (McCoy et al., 2021), plant material characterization (Jaskowska-Lemanska and Przesmycka, 2021), detection and transport of solutes or components, and root structure and development in soil.

5.1. Plant anatomy and analysis of morphological structures

CT has shown significant advantages for anatomical and analytical studies of plant morphology and structure because of its non-invasive detection properties. X-ray CT allows the reconstruction of the 3D geometry of flowers and the visualization of voids in the reproductive organs of plants (Reich et al., 2020), which have been used to gain insight into the anatomy and internal organization of plant fossils, including the "monkey hair tree" in German lignite (McCoy et al., 2021) and the fossilized fruits of the American genus Mauve (Manchester et al., 2021). Regarding the anatomical structure of Eucalyptus dry wood, quantitative and qualitative data on pits, thin-walled tissue, and others, can be obtained in X-ray CT images (Ramos et al., 2020). Murai et al. (2021) focused on the internal structural changes of wood samples during pyrolysis and observed interstices within the wood. Similarly, CT has been used to nondestructively study the tissue structure of plant, such as stem nodes and tree rings (Torres-Silva et al., 2020), extending the study of plant pest damage (Schmidt et al., 2020). Vaz et al. (2020) used MRI, X-ray CT, and μCT to investigate symptoms of grapevine trunk defects due to fungal infections. They showed that high resolution CT is more advantageous in nondestructive identification. However, it has also been argued that CT is not suitable for in situ testing of live trees (Taskhiri et al., 2020b), because it takes time to capture from multiple directions. The reconstruction of the image is challenging if the sample under test changs as it is difficult for CT to capture motion images.

Automatic detection techniques for wood defects are key to optimizing the value of forest products in manufacturing processes (Beaulieu and Dutilleul, 2019). Typically, the options to detect internal defects in trees include ultrasound techniques (Taskhiri et al., 2020b), three-dimensional stress wave imaging method (Du et al., 2018) and synchrotron X-ray CT (Murai et al., 2021; Singh et al., 2020b). However, Saeidi et al. (2020) considered that acoustic waves cannot propagate along a straight line in heterogeneous materials (e.g., oil palm trunks), making it harder for ultrasound to locate cavities in the xylem. High-resolution CT is limited by problems of size and distance, γ-ray CT is too bulky for the equipment. They recommended microwave tomography as it works by identifying the dielectric properties of the material and is usually more suitable for identifying cavities. Given that plants are tissues with complex multicellular structures, Zhang et al. (2020) developed a program for cell identification and quantification, suitable for analyzing the global nature of cells.

In addition, CT has great potential in the field of grain and fruit science. A detailed spatial description of the structural damage of seeds at different fertility stages could be provided with X-ray μCT visualization tools (Zhou et al., 2020). Su and Xiao (2020) provided a solution for the identification and screening of chalky phenotypes in living rice seeds with High Resolution μCT. Van De Looverbosch et al. (2020) proposed an automatic identification and quality grading method for fruits using translational X-ray CT, which can later be applied for rapid sorting of fruits. Texture analysis of X-ray μCT images has also been shown to distinguish rice grains from rice blast infestation (Srivastava et al., 2020), and Synchrotron micro-X-ray fluorescence (micro-XRF) technique has been used to detect trace elements (iron and zinc) in root, shoot, and grain sections in situ (Guo et al., 2022), which has far-reaching implications for grain development and quality control.

5.2. Plant material properties quantification

Because the intensity of CT data or image property highly correlates with the density of the tissue material (Helgason et al., 2008), researchers have been inspired to apply CT scanning to quantify the properties of plant tissues (Kurei et al., 2021; Tao et al., 2020). For example, synchrotron μCT was used to quantify the nanoscale deformation of individual wood cells (Sanabria et al., 2020) and to determine the impact resistance of materials in the loading direction by segmenting different types of cells by tracking their microscopic geometric changes during mechanical loading (e.g., tensile or compressive) (Matsushita et al., 2020). 3D imaging of X-ray μCT scanning has been used to reflect the true structure (Liu et al., 2020) and variation in the stem internodes of rice in order to identify important influences on collapse resistance in rice. Stubbs et al. (2020) developed a method to determine the X-ray CT strength versus material transverse elastic modulus to assess the spatial distribution of material properties of corn stover materials. The structural changes inside the test sample recorded by CT together with image analysis and model building (Li et al., 2020b) enable the analysis of stress characteristics, elasticity (Stubbs et al., 2020), and shear modulus and viscoelasticity under the microstructure of real materials, which is less possible for regular methods such as bio-electron microscopy (Wang et al., 2020a).

5.3. Presence and transport of substances in plants

Researchers have used CT image values to estimate the water content of wood (Yasushi, 1992) and to determine the density of wood (Couceiro et al., 2020; Wang et al., 2020b). Ossler et al. (2021) visualized the spatial distribution of elements H\C\O in wood of different tree species and their biochar formation using both ex-situ neutron computed tomography (NCT) and X-ray CT, but its resolution is still not desirable. The development of near-field X-ray holographic nano-tomography (XNH) (Schudel et al., 2021) can scan at very high resolution to visualize tiny structures (e.g., depressed membranes, cell wall surface structures). McElrone et al. (2021) used this method to demonstrate that water flows primarily in discrete xylem along the axis of the tree trunk. The pathway of the modifier into the modified wood can also be obtained from X-ray μCT analysis (Softje et al., 2020), but the results may be resolution dependent. As with soil, the addition of tracers is a routine method for visualizing flow. Wascher et al. (2020) added an iodine contrast agent to visualize the distribution and flow path of melamine resin in beech veneer. Bubba et al. (2020) treated time as the third dimension and proposed a method for reconstructing images of moving objects from sparse dynamic data, which is responsive for capturing the motion transport process of perfused fluids such as contrast agents. In addition, CT was also used to detect and analyze the activity (Choi et al., 2017; Himmi et al., 2016; Keszthelyi et al., 2016) and distribution (Robin et al., 2018; Teixeira-Costa and Ceccantini, 2016) of animals or microorganisms (Lewinska et al., 2016) within plants.

5.4. Study of root-soil integration

5.4.1. Root system visualization

Understanding the root system structure and root growth state is essential for plant productivity. In the studies of plant root system architecture (RSA) and root growth processes in soil, the opacity of the soil body affects direct visualization of root (Mooney et al., 2012). However, conventional methods such as excavation and profiling are destructive and labor-intensive and CT becomes a powerful analytical tool (Metzner et al., 2015; Rogers et al., 2016; Tumlinson et al., 2008). Anderson et al. (1994) pioneered the use of X-ray CT non-destructive detection techniques in root studies.

However, CT sampling may underestimate root length (Perret et al., 2007), because fine roots below instrumental resolution and therefore large number of small roots near the main root cannot be distinguished from the soil substrate (Fang et al., 2019a; Heeraman et al., 1997). It is tricky for scientists to distinguish soil components (e.g., pore water and organic matter) from plant roots (Karunakaran et al., 2015; Yang et al., 2017) as it shows similar signal as that of organic matter, although the significance of plant roots is widely acknowledged (Zhao et al., 2020). There are also researchers following the medical treatment using contrast agents (Keyes et al., 2017), but it is complicated to estimate the time and dose of contrast agents reaching the root system through the plant. There are potential hazards of contrast agents for living plant bodies (excessive toxicity) that limit its applications (Scotson et al., 2021b).

Another issue is that different root systems show similar grayscale values in CT images because of similar attenuation coefficients due to density, making them difficult to be identified separately. Mairhofer et al. (2015) used root cross-sections in combination with X-ray μCT data to describe the three-dimensional framework of multiple interacting roots in the soil. Although their method was able to isolate the roots of different plants, the explanation for multi-root competition has yet to be broken through. Chirol et al. (2021) developed a hybrid segmentation method to visualize and quantify the spatial analysis of wetland roots and enhanced tubular structures with a "Frangi" filter to distinguish living roots from necrotic tissue, but the segmentation method has yet to gain widespread acceptance. The root segmentation algorithm "Rootine" (Gao et al., 2019b) and the subsequent version "Rootine V.2" (Phalempin et al., 2021) was developed with higher detection accuracy, but segmentation of root-stubble-soil complexes is still a big challenge. Zhao et al. (2020) performed 3D modeling of maize root stubble by segmenting roots and stubble by means of variational level sets, but their method was not very effective in extracting multiple interacting lateral roots. The scan-reconstruction-segmentation process takes a long time because there are a lot of interactive parameter interpretation and optimization (Pfeifer et al., 2015), while manual segmentation is time-consuming and error-prone, requiring the development of an efficient and automated process (Gerth et al., 2021; Teramoto et al., 2020).

5.4.2. Rhizosphere application

The interaction between the soil and the root system plays a key role in agro-environment. Compared to destructive methods such as digging profiles and rinsing roots, CT provides insight into this interaction. On the one hand, both the physical structure and chemical composition, e.g., nutrients (Gao et al., 2019a), of the soil influence root growth. In the X-ray μCT images, there was a strong positive correlation between root volume and the surface/volume ratio of solids, which means that roots grew better in porous soil and the volume and surface of root decreased in sites with high land use intensity (Kuka et al., 2013). Root diameter increased under soil compaction conditions, accompanied by swelling of the root tip (Tracy et al., 2012). Later, Tracy et al. (2013) explored the effects of bulk density and texture on crop root development. Their results stressed that it is important to consider specific time points, because the greater effect of bulk density on root system was manifested at day 3, while the main factor became texture at day 10. In addition, Blaser et al. (2020) used micro suction cups and X-ray μCT to monitor the dynamics of soil nutrients and the response of in situ root morphology. Their results indicated that high NO_3 inhibited the growth of transverse roots and high NH_4^+ stimulated the formation of primary lateral roots. In turn, root development can have an impact on the soil (Galdos et al., 2020). In some cases plant roots can make the soil denser (Helliwell et al., 2019), which significantly reduces the water retention capacity of the inter-rooted soil (Daly et al., 2015). In contrast, Hu et al. (2019) reported that the developed root system under shrub encroachment resulted in increased soil macroporosity and connectivity. Floriana et al. (2021) showed that the soil response to plant root growth is related to the initial soil capacity, such that this response shows swelling characteristics in denser soils and shows a slight shrinkage in looser soils. In addition, the root system also leads to chemical changes in the inter-root soil. For instance, the roots emit a variety of organic compounds that significantly increase microbial activity and dissolve iron and sulfur elements near the root system (Veelen et al., 2020).

Roots can actively detect pore spaces in the soil (Pfeifer et al., 2014). The root-soil interface is unique and critical, but it is not easy to determine the soil-root contact. Schmidt et al. (2012) demonstrated that X-ray μCT can determine the root-soil contact area in situ with an accuracy of ± 3 %. Keyes et al. (2013) used Synchrotron Radiation X-ray Tomographic Microscopy (SRXTM) to show live root hairs of wheat and to assess root hair uptake of phosphorus. Daly et al. (2016) modeled nutrient uptake by root hairs based on structural X-ray CT images of inter-root soil. It is also noted an increasing volume of studies on inter-root bacterial microbial life activities (Ganther et al., 2020) and inter-root soil solution organic matter (Lohse et al., 2020; Mawodza et al., 2020).

Although CT scanning is considered a nondestructive detection technique that has little effect on soil and water properties with prescribed radiation dose, it has negative effects on plants (e.g., plant mutations in breeding) and organisms in the soil, especially at high doses of X-ray or γ-ray exposure (Magdy et al., 2020; Pérez-Jiménez et al., 2020). But these radiation applied doses have not been measured in most studies, but rather estimated by a RadPro calculator (v3.26). Lippold et al. (2021) used actual measurements from plant growth experiments to test the estimates from the calculator and concluded that the calculator was only suitable for assessing attenuation in air. In addition, the use of appropriate radiation doses in plant growth experiments was requested, and they suggested that the critical dose for plant growth barriers in a typical single pot scan was 1.2 Gy.

6. Conclusions and perspectives

The use of CT analysis techniques as a non-invasive analytical tool in soil and plant science has been reported extensively. The basics of CT and image processing are briefly presented. A bibliometric analysis was performed to demonstrate the increasing trend on CT publications and identify the research hotspots. CT applications in plant and soil science were reviewed and discussed. The knowledge gaps and perspectives for future study are:

(1) We must be aware that CT analysis is mainly used as a means to assess the structural characteristics of soils and plants, and that it has some shortcomings, such as the dynamics of solute transport, where the combination of complementary techniques (fluorescence tracing, nuclear magnetic resonance, etc.) with other techniques has become a trend. In future research, it is possible to accelerate the construction and application of 4D-CT technology in soil and plant science research, which will be very friendly for the study of integrated disciplinary problems such as root-soil interaction and interfacial exchange, as well as the dynamic measurement of solute transport in soil and plants (e.g., transport and distribution of heavy metals in super-enriched plants), but new 4D image analysis procedures needs to be developed (Ferreira et al., 2022b). Moreover, for studies related to plants and organisms, the use of radiation doses needs to be more careful.

(2) CT instruments cannot distinguish liquid water, ice, and freezing front in frozen soil, which limits the application of CT technology in frozen soils related studies. However, its combination with MRI or NMR technology that can detect change in unfrozen water content is promising to study freezing and thawing processes. Methods like this combination of different technologies require multiple modality imaging that can combine 2D and 3D data from multiple acquisition modes. It offers new perspectives across scales/dimensions/technologies compared to the traditional "single-mode" approach, and is likely to be the most popular imaging method in the future as it can identify features that cannot be distinguished in a single dimension (Lucas et al., 2020; Mitchell et al., 2021).

(3) For the study of soil health, it is necessary to strengthen the application of CT to promote the understanding of soil health process from the micro scale. This would give insights into the role of microorganisms in soil carbon turnover and sequestration, the formation of soil health and the characterization of the spatio-temporal evolution of biological communities, the efficient utilization of soil nutrients, and the environmental remediation of contaminated soil.

(4) In an added bonus, a long-standing dilemma in CT scanning is the relationship between resolution and sample size and scan size. And limited by the resolution obtained by imaging tools, a large proportion of the sub-resolution pore space in soils are tentatively unidentifiable, where are important habitats for bacteria, archaea, and viruses in soils (Pot et al., 2022a). Therefore, accurate imaging analysis of sub-resolution pore Spaces may provide new insights into soil microhabitats and microbial activity, but this does not sound like something that we shall be able to achieve any time soon. More realistically, we should ensure that the resolution of the CT images is as fine as needs to be. Efforts to develop a new generation of synchrotron quality X-ray sources may allow us to look forward to the future use of relatively inexpensive monochromatic X-ray beams, which will be of great benefit in future research. For image reconstruction and processing analysis, there are different interpretations at different resolutions, and it is difficult to compare between different means of use, so we suggest that this needs a global map network to distinguish and compare image processing tools and algorithms used by scientists worldwide when applying CT in soil or plant science research, to summarize commonalities and advantages, to accelerate the establishment of standards, and that this map needs to be kept up to date. For modeling, a new generation of mathematical models should be developed to better derive information from CT images, For instance, the introduction of dynamic soil structure in micro-scale modeling for soil microorganisms (Pot et al., 2022b), more in-depth modeling and analysis of soil fluid movement and root competition effects.

(5) For equipment, we summarized the different CT types and different energy sources used in several popular application areas, and compared their limitations (Table 2). In general, CT scanning based on X-ray radiation is still the mainstream. Due to the complementary attenuation characteristics of neutron and X-ray, it is expected that their combined method will be applied in a large number of fields. The future may belong to more accurate tabletop multicolor CT or dedicated synchrotron monochrome X-ray beam CT, owing to the call for high resolution and dedicated detection of specific substances. Dealing with the artifacts caused by beam hardening is an urgent problem to be solved in most desktop X-ray CT devices, and more parameters may bring more subjective choices. The benefits of synchrotron are well established. X-ray produced by synchrotron has the characteristics of high intensity, high collimation and tunability, which can effectively avoid the noise problem caused by scattered photons and the beam hardening effect related to multicolor beams (Lavrukhin et al., 2021). If routine access to synchrotrons can be achieved in the future as technology advances, then dedicated synchrotron monochrome X-ray beam CT may be more welcome, especially for specific substances or elements (Aidene et al., 2021; Guo et al., 2022).

Finally, it is expected that CT scanning would have great opportunity for future development in the directions of soil health and soil management, dynamic study of solute transport and visual presentation of

Table 2
Methods and energy types used in different applications in recent years.

Application	Category	Energy	Advantages/ Limitations
Structure/ pore	X-ray CT (Schluter et al., 2020)	X-ray	Ubiquitous; device access is relatively easy; limited resolution; artifacts caused by multicolor beams
	Dual-resolution X-ray CT (Zeng et al., 2022)	X-ray	Limited by sample size
	X-ray micro-computed tomography (Yu et al., 2021)	X-ray	Resolution is higher than ordinary X-ray CT; micron level; small sample size
	SR-μCT (Ma et al., 2021)	X-ray	High resolution; high collimation; monochrome; micron level; global device access is difficult
	Dual energy procedure (X, γ) rays CT (Fonseca et al., 2019)	X-ray and γ-ray	Improve the effectiveness of X-ray CT; high precision representation; γ-ray CT requires a good collimated narrow beam; time consuming
Organic matter/ organism	Dual-energy X-ray μCT (staining assisted) (Zheng et al., 2020)	X-ray	Limited by sample size; limited by dyeing effect; pay attention to the toxicity
	SR-μCT (Lammel et al., 2019)	X-ray	Same as above; can reveal local mycelial morphology
	Joint X-ray and time-of-flight (TOF) neutron imaging (Koestel et al., 2022)	X-ray and neutron	Taking advantage of the complementary attenuation characteristics of neutrons and X-rays; inability to distinguish SOM from clay minerals (at voxel scale)
	X-ray CT (Cheik et al., 2019)	X-ray	Same as above
	μX-ray fluorescence microscopy (Védère et al., 2022)	Fluorescent radiation excited by an X-ray beam	Problem of fluorescence attenuation needs to be avoided
Water, gas, solutes (heavy metal elements)	(Dual energy) X-ray CT (contrast medium) (Guber et al., 2021)	X-ray	Relatively common; increases the difficulty of segmentation; difficult to identify heavy elements with atomic numbers much higher than about 40
	Triple-Exposure X-ray CT (Nakashima, 2021)	X-ray	Heavy elements in clay minerals can be identified; complicated operation
	Neutron tomography (Jailin et al., 2018)	Neutron	High sensitivity to light elements; high penetration; little radiation damage; high cost; high time consuming; weak neutron source; small number of available instruments
	ERT(Chaali et al., 2022)	Conductance	Monitor fluid flow; large-scale; two-dimensional measurements; cannot be used in the presence of conductive materials (e.g. clay); surface conductivity is often neglected
	Spectral induced polarization (SIP) tomography (Revil et al., 2021)	Conductance	Coarse elements (with very low relaxation frequencies) were excluded; the quality of the forecasts needs to be further evaluated
	Spectral computed tomography (Sittner et al., 2021)	X-ray (absorption spectra)	High resolution; can identify minerals and specific elements (such as heavy elements) in materials; artifact problem caused by multi-color light needs to be solved
	Micro X-ray fluorescence (Gattullo et al., 2020)	Fluorescent radiation excited by an X-ray beam	Problem of fluorescence attenuation needs to be avoided
Plants/roots	Microwave tomography (Saeidi et al., 2020)	Electromagnetic wave	Safe; low cost; easy to use; limited resolution
	ultrasonic tomography (Taskhiri et al., 2020a)	Ultrasonic	No radiation damage; limited resolution
	ERT (Giambastiani et al., 2022)	Conductance	Same as above
	X-ray (μ) CT (Hou et al., 2022)	X-ray	Subject to sample size; difficult to distinguish root and soil organic matter
	SR-μCT (Hou et al., 2022)	X-ray	Same as above
	Positron emission tomography (PET)/CT (Garbout et al., 2012)	Electronic, X-ray	Suitable for larger samples; fewer cases in recent years
	Neutron tomography (Mawodza et al., 2020)	Neutron	Same as above; can visualize the roots in soils with high organic matter content; complexity of segmentation

freezing and thawing processes in frozen soils, which is expected to provide a reference for the promotion of CT technology in agriculture and environment research.

CRediT authorship contribution statement

Conceptualization, **H.H. and L.Y.**; Methodology, **H.Z.**; Software, **H.Z.**; Validation, **J.L.**; Formal analysis, **H.Z.**; Investigation, **H.Z.**; resources, **H.F.**; Data curation, **H.Z.**; Writing – original draft preparation, **H.Z.** and **H.H.**; Writing – review & editing, **H.H., A.M., M.D.,** and **V.F.**; Visualization, **H.Z.** and **Y.G.**; Supervision, **H.H.** and **L.Y.**; Project administration, **H.H.**; Funding acquisition, **H.H.**; All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article. Scientometric data were retrieved from the Web of Science Core Collection with query sets presented in the method and material section. The download datafile is available upon request.

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Consent to participate/Ethical approval

All authors agree to participate and publish.

Animal research

Not applicable.

Consent to publish

Not applicable.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.still.2022.105574.

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